

Stat 2300 International, Fall 2006 – Final Answers

1. **c.** Less than α
2. **d.** Whether a Person Has a Traffic Violation
3. **a.** failing to reject, be rejected
4. **b.** $H_0: \mu \leq 22$ vs. $H_a: \mu > 22$
5. **b.** Union
6. **a.** $\mu_{\text{consumer}} - \mu_{\text{banking}} \leq 0$
7. **b.** t statistic
8. **d.** 0 and 1
9. **c.** 40
10. **b.** the same
11. **a.** Positive autocorrelation of error terms
12. **b.** 2
13. **d.** 48 and 8
14. **b.** .5604
15. **c.** 0.580
16. **a.** A prediction based on Model I is likely better than a prediction based on Model II.
17. **a.** -1.2024
18. **b.** t test
19. **a.** $\frac{n!}{x!(n-x)!}$
20. **c.** Narrower for 90% confidence than for 95% confidence.

21. Answer: .1818

$$P(B \cap E) = (.20)(.05) = .01$$

$$P(E) = (.45)(.10) + (.20)(.05) + (.35)(0) = .055$$

$$P(B|E) = \frac{P(B \cap E)}{P(E)} = \frac{.01}{.055} = .1818$$

22. Answer: Expected Profit = $0.01 * (-19,500) + 0.99 * 500 = \300

23. Answer:

$$\mu = np = 500 * .6 = 300, \quad \sigma = \sqrt{np(1-p)} = \sqrt{500 * .6 * .4} = \sqrt{120} = 10.95$$

$$P(X = 305) = P(304.5 \leq X \leq 305.5) =$$

$$P((304.5 - 300)/10.95 \leq Z \leq (305.5 - 300)/10.95) =$$

$$P(.41 \leq Z \leq .50) =$$

$$.6915 - .6591 = .0324 = 3.24\%$$

24. Answer:

Mean	466.25
Confidence Level(90.0%)	39.9136996029933
90% CI:	[426.3363, 506.1637]

25. Answer:

(i)

Mean 63.4

(ii)

Median 49.6

(iii)

[Sample Variance 1255.90666666667] =>

Population Variance = $9/10 * 1255.90666666667 = 1130.316$

26. Answer: $H_0: \mu \leq 50$ vs. $H_A: \mu > 50$

27. Answer: $t = (54 - 50) / (8 / \sqrt{25}) = 2.5$

28. Answer: about 2.492, based on a t distribution with $df = 24$ (as n is less than 29)

29. Answer: $\bar{x} \pm t_{n-1, \alpha/2} * \frac{s}{\sqrt{n}} = 54 \pm 2.797 * \frac{8}{\sqrt{25}} = [49.52, 58.48]$

30. Answer: Intercept (b_0) = 14.86764802; Slope (b_1) = 0.937119568

31. Answer:

$$\hat{y} = b_0 + b_1 * x = 14.86764802 + 0.937119568 * 200 = 202.292 \text{ (i.e., \$202,292)}$$

32. Answer:

$$SSE = 2467.807; s^2 = SSE / (n-2) = 2467.807 / 13 = 189.8312851;$$

$$s = \sqrt{s^2} = 13.77792746$$

33. Answer:

$$s_{b_1} = 0.073045; t = 12.82927; \text{ this is calculated as:}$$

$$t = \frac{b_1}{s_{b_1}} = 0.937119568 / 0.073045 = 12.82927$$

34. Answer:

From Excel output: p -value = 9.34E-09; reject the null hypothesis for all given values of alpha; there is an extremely strong evidence of a significant relationship between number of households and electronic sales.

35. Answer:

From Excel output: [Lower 90%, Upper 90%] = [0.807760954, 1.066478182]

Note: We are not looking for [Lower 95%, Upper 95%] = [0.779314483, 1.094925]

36. Answer: Supermarket 1: 24; 2: 21.66667; 3: 9.333333; 4: 13.66667

37. Answer:

(i) 959.1818 (ii) 2 (iii) 3.03E-08 (iv) reject H_0 at $\alpha = .05$ (v) different bottle designs DO have different effects on mean sales

38. Answer:

(i) 460 (ii) 3 (iii) 1.77E-07 (iv) reject H_0 at $\alpha = .05$ (v) different supermarkets DO have different effects on mean sales

39. Answer:

$p = 3$ treatments, $b = 4$ blocks; (i) $r = p = 3$ (ii) $v = (p-1)(b-1) = 2*3 = 6$ (iii) for 95% we need $q_{.05}$ (iv) value of $q_{.05} = 4.34$

40. Answer:

General: $(\bar{x}_{i\bullet} - \bar{x}_{j\bullet}) \pm q_\alpha * \frac{s}{\sqrt{b}}$ where $s = \sqrt{MSE} = \sqrt{0.305556} = 0.552771$

AB: $9.25 - 26.25 \pm 4.34 * \sqrt{0.305556} / \sqrt{4} = [-18.1995, -15.8005]$

AC: $9.25 - 16 \pm 4.34 * \sqrt{0.305556} / \sqrt{4} = [-7.94951, -5.55049]$

BC: $26.25 - 16 \pm 4.34 * \sqrt{0.305556} / \sqrt{4} = [9.050487, 11.44951]$

As none of these intervals contains zero, we can conclude that bottle design B maximizes sales.

Intervals with $q_\alpha = 4.0$:

AB: $[-18.1055, -15.8945]$

AC: $[-7.85554, -5.64446]$

BC: $[9.144458, 11.35554]$